Software Requirements Specification for Software Engineering: subtitle describing software

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Contents

1 Purpose of the Project **vi**
- 1.1 User Business . . . vi
- 1.2 Goals of the Project . . . vi

2 Stakeholders **vi**
- 2.1 Client . . . vi
- 2.2 Customer . . . vi
- 2.3 Other Stakeholders . . . vi
- 2.4 Hands-On Users of the Project . . . vi
- 2.5 Personas . . . vi
- 2.6 Priorities Assigned to Users . . . vi
- 2.7 User Participation . . . vii
- 2.8 Maintenance Users and Service Technicians . . . vii

3 Mandated Constraints **vii**
- 3.1 Solution Constraints . . . vii
- 3.2 Implementation Environment of the Current System . . . vii
- 3.3 Partner or Collaborative Applications . . . vii
- 3.4 Off-the-Shelf Software . . . vii
- 3.5 Anticipated Workplace Environment . . . vii
- 3.6 Schedule Constraints . . . vii
- 3.7 Budget Constraints . . . vii
- 3.8 Enterprise Constraints . . . viii

4 Naming Conventions and Terminology **viii**
- 4.1 Glossary of All Terms, Including Acronyms, Used by Stakeholders involved in the Project . . . viii

5 Relevant Facts And Assumptions **viii**
- 5.1 Relevant Facts . . . viii
- 5.2 Business Rules . . . viii
- 5.3 Assumptions . . . viii

6 The Scope of the Work **viii**
- 6.1 The Current Situation . . . viii
- 6.2 The Context of the Work . . . viii
- 6.3 Work Partitioning . . . ix

6.4 Specifying a Business Use Case (BUC) ix

7 Business Data Model and Data Dictionary **ix**
7.1 Business Data Model ix
7.2 Data Dictionary ix

8 The Scope of the Product **ix**
8.1 Product Boundary ix
8.2 Product Use Case Table ix
8.3 Individual Product Use Cases (PUC's) ix

9 Functional Requirements **ix**
9.1 Functional Requirements ix

10 Look and Feel Requirements **x**
10.1 Appearance Requirements x
10.2 Style Requirements x

11 Usability and Humanity Requirements **x**
11.1 Ease of Use Requirements x
11.2 Personalization and Internationalization Requirements x
11.3 Learning Requirements x
11.4 Understandability and Politeness Requirements x
11.5 Accessibility Requirements x

12 Performance Requirements **x**
12.1 Speed and Latency Requirements x
12.2 Safety-Critical Requirements xi
12.3 Precision or Accuracy Requirements xi
12.4 Robustness or Fault-Tolerance Requirements xi
12.5 Capacity Requirements xi
12.6 Scalability or Extensibility Requirements xi
12.7 Longevity Requirements xi

13 Operational and Environmental Requirements **xi**
13.1 Expected Physical Environment xi
13.2 Wider Environment Requirements xi
13.3 Requirements for Interfacing with Adjacent Systems xii
13.4 Productization Requirements xii

13.5 Release Requirements xii

14 Maintainability and Support Requirements **xii**
14.1 Maintenance Requirements xii
14.2 Supportability Requirements xii
14.3 Adaptability Requirements xii

15 Security Requirements **xii**
15.1 Access Requirements xii
15.2 Integrity Requirements xii
15.3 Privacy Requirements xiii
15.4 Audit Requirements xiii
15.5 Immunity Requirements xiii

16 Cultural Requirements **xiii**
16.1 Cultural Requirements xiii

17 Compliance Requirements **xiii**
17.1 Legal Requirements xiii
17.2 Standards Compliance Requirements xiii

18 Open Issues **xiii**

19 Off-the-Shelf Solutions **xiii**
19.1 Ready-Made Products xiii
19.2 Reusable Components xiv
19.3 Products That Can Be Copied xiv

20 New Problems **xiv**
20.1 Effects on the Current Environment xiv
20.2 Effects on the Installed Systems xiv
20.3 Potential User Problems xiv
20.4 Limitations in the Anticipated Implementation Environment That May Inhibit the New Product xiv
20.5 Follow-Up Problems xiv

21 Tasks **xiv**
21.1 Project Planning xiv
21.2 Planning of the Development Phases xv

22 Migration to the New Product **xv**
22.1 Requirements for Migration to the New Product xv
22.2 Data That Has to be Modified or Translated for the New System xv

23 Costs **xv**

24 User Documentation and Training **xv**
24.1 User Documentation Requirements xv
24.2 Training Requirements xv

25 Waiting Room **xv**

26 Ideas for Solution **xv**

27 Appendix — Generative AI Use Disclosure **xvi**
27.1 AI Tools Used xvi
27.2 How and Where Used xvi
27.3 Verification of AI-Generated Content xvi

28 Appendix — Reflection **xvii**

Revision History

Date	Version	Notes
Date 1	1.0	Notes
Date 2	1.1	Notes

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27 Appendix — Generative AI Use Disclosure

[ChatGPT (version 5) was used to brainstorm ideas for the template for this disclose section. —SS]

27.1 AI Tools Used

[Give the name of the AI tool(s) used and the version. —SS]

27.2 How and Where Used

[Explain what the tool(s) were used for. Examples include: brainstorming, explaining a technical concept, reviewing the document, generating or modifying text, generating or modifying diagrams, identifying errors or omissions or inconsistencies. —SS]

[For each of the uses, identify which parts of the document were affected. —SS]

[You do not need to report every interaction, but you should disclose the uses that materially contributed to your work —SS]

27.3 Verification of AI-Generated Content

[

- Describe how the team checked AI-generated or AI-assisted material before including it in the document.
- In particular, verify factual claims, technical assertions, requirements, calculations, references, and other information for which an AI system may produce incorrect or fabricated results.
- **The team is responsible for the accuracy, completeness, consistency, and appropriateness of the submitted document, regardless of whether material was generated by a human or an AI system.**

—SS]

28 Appendix — Reflection

The purpose of reflection questions is to give you a chance to assess your own learning and that of your group as a whole, and to find ways to improve in the future. Reflection is an important part of the learning process. Reflection is also an essential component of a successful software development process.

Reflections are most interesting and useful when they're honest, even if the stories they tell are imperfect. You will be marked based on your depth of thought and analysis, and not based on the content of the reflections themselves. Thus, for full marks we encourage you to answer openly and honestly and to avoid simply writing "what you think the evaluator wants to hear."

Please answer the following questions. Some questions can be answered on the team level, but where appropriate, each team member should write their own response:

1. What went well while writing this deliverable?
   Before beginning the deliverable, the group discussed each of the required sections together and took detailed notes for each one. This helped with writing the deliverable because everyone in the group was on the same page and understood the approach we wanted to take. This allowed us to easily divide up the writing work and ensure that each section would be cohesive and unified. Additionally, the group set an internal deadline to complete the writing the day before the official deliverable due date. This helped keep us on track and accountable for our work. Additionally, we were able to make some improvmenets on the group's workflow from deliverable 1 and address some pain points experienced for that one.

2. What pain points did you experience during this deliverable, and how did you resolve them?
   Like with the last deliverable, our group had to deal with some technical difficulties. For example, several team members ran into difficulties with GitHub and merge conflicts. We were able to resolve these by working together and using pull requests. We also had some challenges with finding and chooseing the appropriate templates for each deliverable,

as guidance on this was spread across several locations and requirements weren't always clear. We resolved this by sticking to the SRS checklist and askign for clarification where we were unsure.

3. How many of your requirements were inspired by speaking to your client(s) or their proxies (e.g. your peers, stakeholders, potential users)? **Many of the requirements were inspired by conversations with our supervisors. Their input was really valuable in refining our scope and identifying what would actually be useful to an SLP during a session.**

4. Which of the courses you have taken, or are currently taking, will help your team to be successful with your capstone project. **Our Human Computer Interfaces course will help us design an interface that supports the SLP's workflow and that is usable for young children. Our Requirements Engineering course has helped us write clear requirements and will continue to help us trace them through the project. Our Software Architecture course will help us with the structure of the system and design decisions.**

5. What knowledge and skills will the team collectively need to acquire to successfully complete this capstone project? Examples of possible knowledge to acquire include domain specific knowledge from the domain of your application, or software engineering knowledge, mechatronics knowledge or computer science knowledge. Skills may be related to technology, or writing, or presentation, or team management, etc. You should look to identify at least one item for each team member. **The team will need to acquire the following knowledge and skills:**
   Phonetics and child speech challenges, speech analysis models, evaluaiton of speech analysis accuracy, healthcare privacy and secure data handling, and audio capture. The team will split these 5 skills areas amongst each other.

6. For each of the knowledge areas and skills identified in the previous question, what are at least two approaches to acquiring the knowledge or mastering the skill? Of the identified approaches, which will

each team member pursue, and why did they make this choice? **For phonetics and child speech challenges, we will ask our supervisors for details on the international pheonetic alphabet and research introductory phoentics information. For speech analysis models, we will work with existing models and documentation to build our knowledge and research other projects that have been done on speech analysis. For evaluation of speech analysis accuracy, we will study existing benchmarks for the models and run our own tests using them. For healthcare privacy, we will read documentation on PHIPA regulations and consult our supervisors. For audio capture, we will test mockups and consult with our supervisors.**